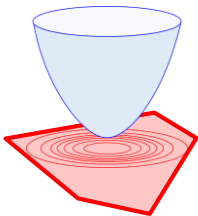


Introduction to Optimization

MATH-224 (OPT): Lecture 3



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Slide Credit: [Xinyu Chen](#)

Linear Programming

- **Concept:** Linear programming or linear optimization
- A special case of mathematical programming
- **Definition:** Linear programming is a technique for the optimization of a **linear objective function**, subject to (s.t.) **linear equality and linear inequality constraints**.
- **Description:** Find the **best outcome** (such as maximum profit or lowest cost) in a model where requirements are represented by **linear** relationships.

Mathematical Programming

- A general formula:

$$\begin{array}{ll} \min_x & f(x) \\ \text{s.t.} & x \in \mathcal{C} \end{array} \quad \text{or} \quad \begin{array}{ll} \max_x & f(x) \\ \text{s.t.} & x \in \mathcal{C} \end{array}$$

- Main components:

- Decision variable x
- Objective function $f(x)$
- Linear constraint $x \in \mathcal{C}$
- Feasible set $\mathcal{C}$, e.g., $\mathcal{C} = \{x \mid x \geq 0\}$

- Format: Minimize (**min**) vs. Maximize (**max**)

Linear Programming

- **Example:**

$$\begin{aligned}
 \max_{x_1, x_2} \quad & c_1 x_1 + c_2 x_2 && \text{(linear objective)} \\
 \text{s.t.} \quad & a_{11} x_1 + a_{12} x_2 \leq b_1 && \text{(linear inequality)} \\
 & a_{21} x_1 + a_{22} x_2 \leq b_2 && \text{(linear inequality)} \\
 & x_1, x_2 \geq 0 && \text{(non-negativity)}
 \end{aligned}$$

or

$$\begin{aligned}
 \max_{\mathbf{x}} \quad & \mathbf{c}^\top \mathbf{x} && \text{(linear objective)} \\
 \text{s.t.} \quad & \mathbf{A} \mathbf{x} \leq \mathbf{b} && \text{(linear inequality)} \\
 & \mathbf{x} \geq 0 && \text{(non-negativity)}
 \end{aligned}$$

for any matrix $\mathbf{A}$ and vectors $\mathbf{b}, \mathbf{c}$.

Geometry of Linear Programming

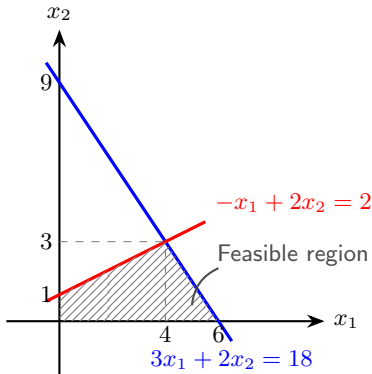
Linear programming:

$$\begin{array}{ll}\max_{x_1 \in \mathbb{R}, x_2 \in \mathbb{R}} & x_1 + x_2 \\ \text{s.t.} & 3x_1 + 2x_2 \leq 18 \\ & -x_1 + 2x_2 \leq 2 \\ & x_1, x_2 \geq 0\end{array}$$

Geometry of Linear Programming

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Linear constraints define the **feasible region**, which is a convex polytope.

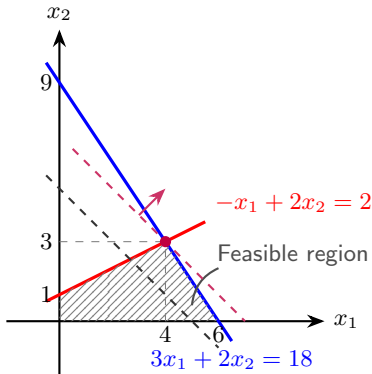
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Optimal solution:

$$x_1^* = 4 \quad x_2^* = 3$$



Linear constraints define the **feasible region**, which is a convex polytope.

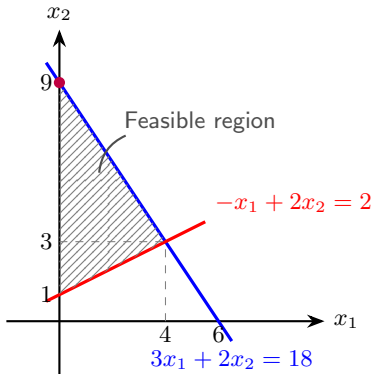
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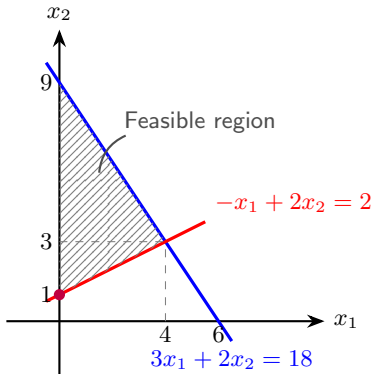
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Optimal solution:

$$x_1^* = 0 \quad x_2^* = 1$$



Linear constraints define the **feasible region**, which is a convex polytope.

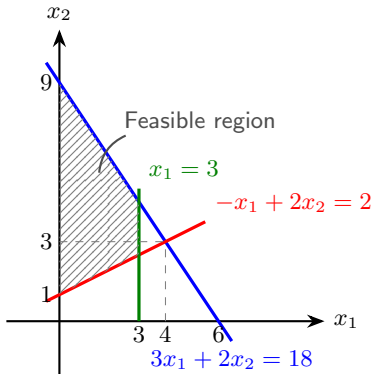
Geometry of Linear Programming

Linear programming:

$$\begin{aligned}
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 \end{aligned}$$

Optimal solution:

$$x_1^* = 3 \quad x_2^* = 4.5$$



Linear constraints define the **feasible region**, which is a convex polytope.

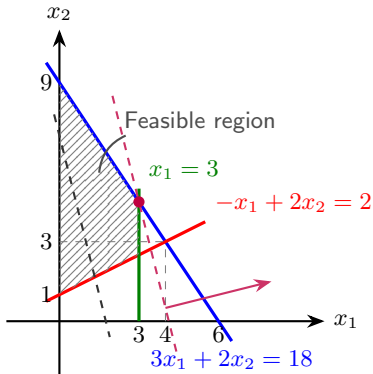
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Linear programming:

$$\begin{aligned}
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Optimal solution:

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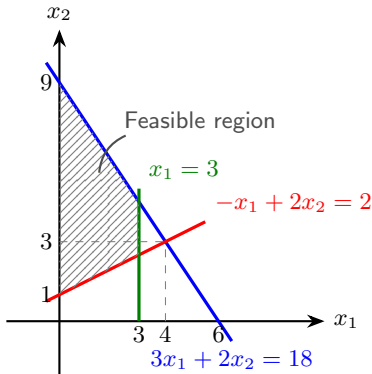
Linear constraints define the **feasible region**, which is a convex polytope.

Geometry of Linear Programming

Linear programming:

$$\begin{aligned}
 \min_{x_1 \in \mathbb{R}, x_2 \in \mathbb{R}} \quad & -x_1 + x_2 \\
 \text{s.t.} \quad & x_1 \leq 3 \\
 & 3x_1 + 2x_2 \leq 18 \\
 & -x_1 + 2x_2 \geq 2 \\
 & x_1, x_2 \geq 0
 \end{aligned}$$

Optimal solution?



Linear constraints define the **feasible region**, which is a convex polytope.

Python Programming

```
1 import cvxpy as cp
2
3 # Define the decision variables
4 x = cp.Variable(2, nonneg=True)
5 # Define the objective function
6 objective = cp.Minimize(-x[0] + x[1])
7
8 # Define the constraints
9 constraints = [
10     x[0] <= 3,                # x1 <= 3
11     3*x[0] + 2*x[1] <= 18,    # 3x1 + 2x2 <= 18
12     -x[0] + 2*x[1] >= 2       # -x1 + 2x2 >= 2
13 ]
14
15 # Formulate and solve the problem
16 prob = cp.Problem(objective, constraints)
17 result = prob.solve()
18 # Output the results
19 print(f'Status: {prob.status}')
20 print(f'Optimal Value (max objective): {prob.value:.4f}')
21 print(f'Optimal x1: {x.value[0]:.4f}')
22 print(f'Optimal x2: {x.value[1]:.4f}')
```

ℓ_1 -Norm Optimization

Given $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$, find the best $x \in \mathbb{R}^n$ that minimizes

$$\varepsilon = b - Ax$$

- The sum of relative residuals:

$$\underbrace{\|\varepsilon\|_1}_{\ell_1\text{-norm}} = \sum_{i=1}^n |\varepsilon_i|$$

- ℓ_1 -norm optimization:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \sum_{i=1}^n |\varepsilon_i| \\ \text{s.t.} \quad & \varepsilon = b - Ax \end{aligned}$$

ℓ_1 -Norm Optimization

- Linear programming with **bound variables** $c_i, i = 1, 2, \dots, n$:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^n, \mathbf{c} \in \mathbb{R}^n} \quad & \sum_{i=1}^n c_i \\ \text{s.t.} \quad & \boldsymbol{\varepsilon} = \mathbf{b} - \mathbf{A}\mathbf{x} \\ & -c_i \leq \varepsilon_i \leq c_i \quad i = 1, 2, \dots, n \\ & c_i \geq 0 \quad i = 1, 2, \dots, n \end{aligned}$$

- Toy example:

$$\mathbf{A} = \begin{bmatrix} 3 & 2 \\ -1 & 2 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 18 \\ 2 \end{bmatrix} \quad \Rightarrow \quad \text{Exact solution: } \mathbf{x} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

ℓ_1 -Norm Optimization

```
1 import cvxpy as cp
2 import numpy as np
3
4 n = 2
5 A = np.array([[3, 2], [-1, 2]])
6 b = np.array([18, 2])
7
8 # Define decision variables
9 x = cp.Variable(n)
10 c = cp.Variable(n, nonneg=True)
11 # Minimize the sum of the bound variables c_i
12 objective = cp.Minimize(cp.sum(c))
13 # Define constraints
14 constraints = [-c <= b - A @ x, b - A @ x <= c]
15 # Formulate and solve the problem
16 prob = cp.Problem(objective, constraints)
17 prob.solve()
18 # Output results
19 print(f'Status: {prob.status}')
20 print(f'Optimal objective value (sum of c): {prob.value}')
21 print(f'Optimal x:\n{x.value}')
```

ℓ_∞ -Norm Optimization

Given $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$, find the best $x \in \mathbb{R}^n$ that minimizes

$$\varepsilon = b - Ax$$

- The worst-case residual:

$$\underbrace{\|\varepsilon\|_\infty}_{\ell_\infty\text{-norm}} = \max\{|\varepsilon_1|, |\varepsilon_2|, \dots, |\varepsilon_n|\}$$

- ℓ_∞ -norm optimization:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \max\{|\varepsilon_1|, |\varepsilon_2|, \dots, |\varepsilon_n|\} \\ \text{s.t.} \quad & \varepsilon = b - Ax \end{aligned}$$

ℓ_∞ -Norm Optimization

- Linear programming with **bound variable c (scalar)**:

$$\begin{aligned}
 \min_{\mathbf{x} \in \mathbb{R}^n, c \in \mathbb{R}} \quad & c \\
 \text{s.t.} \quad & \boldsymbol{\varepsilon} = \mathbf{b} - \mathbf{A}\mathbf{x} \\
 & -c \leq \varepsilon_i \leq c \quad i = 1, 2, \dots, n \\
 & c \geq 0
 \end{aligned}$$

- Toy example:

$$\mathbf{A} = \begin{bmatrix} 3 & 2 \\ -1 & 2 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 18 \\ 2 \end{bmatrix} \quad \Rightarrow \quad \text{Exact solution: } \mathbf{x} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

ℓ_∞ -Norm Optimization

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2 import numpy as np
3
4 n = 2
5 A = np.array([[3, 2], [-1, 2]])
6 b = np.array([18, 2])
7
8 # Define decision variables
9 x = cp.Variable(n)
10 c = cp.Variable(1, nonneg=True)
11 # Minimize the bound variable c
12 objective = cp.Minimize(c)
13 # Define constraints
14 constraints = [-c <= b - A @ x, b - A @ x <= c]
15 # Formulate and solve the problem
16 prob = cp.Problem(objective, constraints)
17 prob.solve()
18 # Output results
19 print(f'Status: {prob.status}')
20 print(f'Optimal objective value: {prob.value}')
21 print(f'Optimal x:\n{x.value}')
```

Quick Summary

- Essential formula of optimization
- What is linear programming?
- Geometric interpretation of linear programming
- Applications to ℓ_1 - and ℓ_∞ -norm optimization

Integer Programming

- **Integer programming** is a mathematical programming in which some or all of the variables are restricted to be **integers**.

Integer decision variable: $x \in \mathbb{Z}$

Binary decision variable: $x \in \{0, 1\}$ (either 0 or 1)

Integer decision vector: $\mathbf{x} \in \mathbb{Z}^n$

- The set of all integers is $\mathbb{Z}$: e.g., $-3, -2, -1, 0, 1, 2, 3$

Geometry of Integer Programming

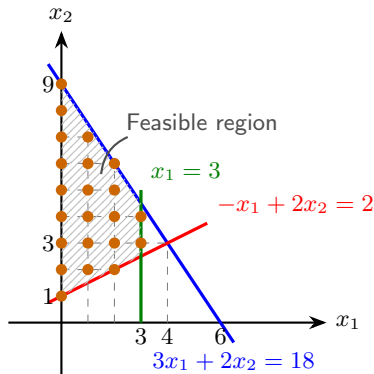
Integer programming:

$$\begin{aligned} \max \quad & 4x_1 + x_2 \\ \text{s.t.} \quad & x_1 \leq 3 \\ & 3x_1 + 2x_2 \leq 18 \\ & -x_1 + 2x_2 \geq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

All feasible solutions (●)

Optimal solution:

$$x_1^* = 3 \quad x_2^* = 4$$



Geometry of Integer Programming

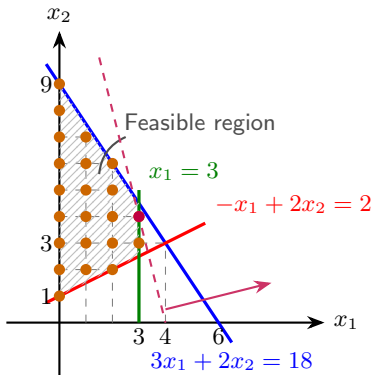
Integer programming:

$$\begin{array}{ll}\max & 4x_1 + x_2 \\ \text{s.t.} & x_1 \in \mathbb{Z}, x_2 \in \mathbb{Z} \\ & x_1 \leq 3 \\ & 3x_1 + 2x_2 \leq 18 \\ & -x_1 + 2x_2 \geq 2 \\ & x_1, x_2 \geq 0\end{array}$$

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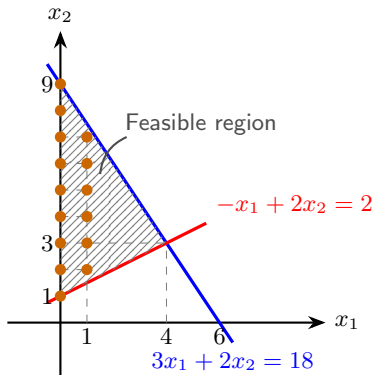
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All feasible solutions (●)

Optimal solution:

$$x_1^* = 1 \quad x_2^* = 7$$



Geometry of Integer Programming

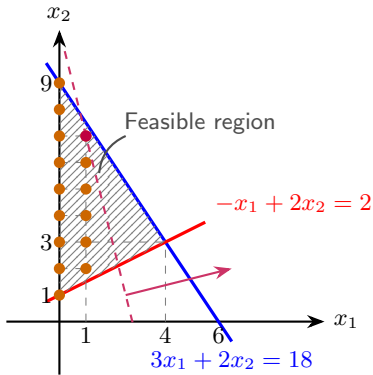
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All feasible solutions (●)

Optimal solution:

$$x_1^* = 1 \quad x_2^* = 7$$



Mixed-Integer Programming

- **Mixed-integer programming** is a mathematical programming that solves problems involving a mix of **continuous** and **integer decision variables**.
 - Continuous decision variables, e.g., $x \in \mathbb{R}$
 - Integer decision variables, e.g., $x \in \mathbb{Z}$, $x \in \{0, 1\}$

Geometry of Mixed-Integer Programming

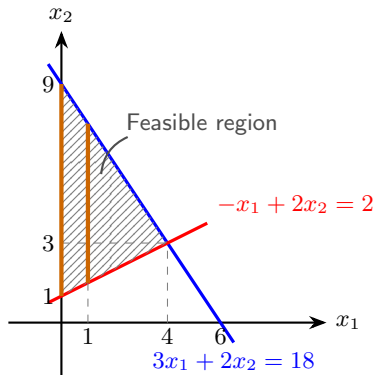
Mixed-integer programming:

$$\begin{aligned} \max \quad & 4x_1 + x_2 \\ \text{s.t.} \quad & 3x_1 + 2x_2 \leq 18 \\ & -x_1 + 2x_2 \geq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

All feasible solutions (—)

Optimal solution:

$$x_1^* = 1 \quad x_2^* = 7.5$$



Geometry of Mixed-Integer Programming

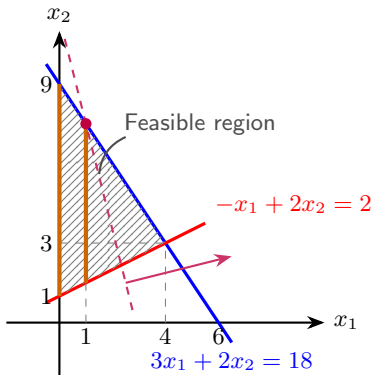
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 \end{array}$$

All feasible solutions (—)

Optimal solution:

$$x_1^* = 1 \quad x_2^* = 7.5$$



Nonlinear Programming

- **Nonlinear programming** is the process of solving an optimization problem where some of the constraints are **not linear equalities** or **the objective function is not a linear function**.

min nonlinear function

s.t. nonlinear equality/inequality

Nonlinear Programming

- **Nonlinear programming** is the process of solving an optimization problem where some of the constraints are **not linear equalities** or **the objective function is not a linear function**.

$$\begin{array}{ll} \min & \text{nonlinear function} \\ \text{s.t.} & \text{nonlinear equality/inequality} \end{array}$$

- **Quadratic programming** is the process of solving certain mathematical optimization problems involving **quadratic functions**.
 - Optimize (minimize/maximize) a multivariate **quadratic function** subject to **linear constraints** on the variables
 - A type of nonlinear programming

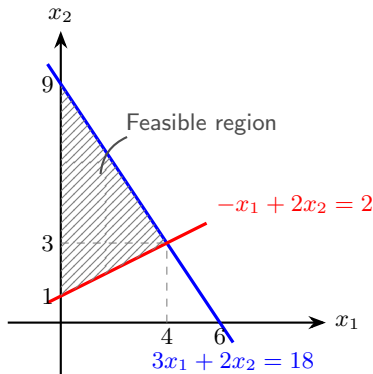
Geometry of Quadratic Programming

Quadratic programming:

$$\begin{aligned} \max_{x_1 \in \mathbb{R}, x_2 \in \mathbb{R}} \quad & x_1^2 + x_2^2 \\ \text{s.t.} \quad & 3x_1 + 2x_2 \leq 18 \\ & -x_1 + 2x_2 \geq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Optimal solution:

$$x_1^* = 0 \quad x_2^* = 9$$



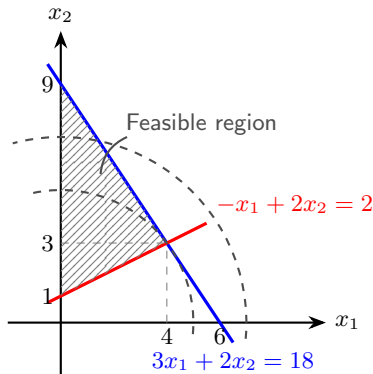
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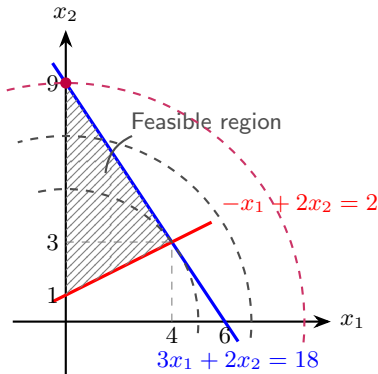
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Reformulating Quadratic Programming

- Reformulate the quadratic programming:

$$\begin{array}{ll}
 \max_{x_1 \in \mathbb{R}, x_2 \in \mathbb{R}} & x_1^2 + x_2^2 \\
 \text{s.t.} & 3x_1 + 2x_2 \leq 18 \\
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 & x_1, x_2 \geq 0
 \end{array}
 \Rightarrow
 \begin{array}{ll}
 \max_{x_1 \in \mathbb{R}, x_2 \in \mathbb{R}} & x_1^2 + x_2^2 \\
 \text{s.t.} & 3x_1 + 2x_2 \leq 18 \\
 & x_1 - 2x_2 \leq -2 \\
 & x_1, x_2 \geq 0
 \end{array}$$

Reformulating Quadratic Programming

- Reformulate the quadratic programming:

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 \max_{x_1 \in \mathbb{R}, x_2 \in \mathbb{R}} & x_1^2 + x_2^2 \\
 \text{s.t.} & 3x_1 + 2x_2 \leq 18 \\
 & -x_1 + 2x_2 \geq 2 \\
 & x_1, x_2 \geq 0
 \end{array}
 \Rightarrow
 \begin{array}{ll}
 \max_{x_1 \in \mathbb{R}, x_2 \in \mathbb{R}} & x_1^2 + x_2^2 \\
 \text{s.t.} & 3x_1 + 2x_2 \leq 18 \\
 & x_1 - 2x_2 \leq -2 \\
 & x_1, x_2 \geq 0
 \end{array}$$

- Let $A = \begin{bmatrix} 3 & 2 \\ 1 & -2 \end{bmatrix}$ and $b = \begin{bmatrix} 18 \\ -2 \end{bmatrix}$
- For decision variables $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2$, we have

$$\begin{array}{ll}
 \max_{x \in \mathbb{R}^2} & \|x\|_2^2 \\
 \text{s.t.} & Ax \leq b \\
 & x \geq 0
 \end{array}$$

with the objective function $\|x\|_2^2 = x^\top x$.

General Form of Quadratic Programming

- Quadratic programming:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \underbrace{x^\top Q x}_{\text{quadratic}} + \underbrace{c^\top x}_{\text{linear}} \\ \text{s.t.} \quad & Ax \leq b \end{aligned}$$

- Symmetric matrix $Q \in \mathbb{R}^{n \times n}$
 - $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$
- In the case of symmetric positive definite Q , it is just a regression!

$$\|\epsilon\|_2^2 = \|b_0 - A_0 x\|_2^2 = \underbrace{x^\top A_0^\top A_0 x}_{\text{quadratic}} - 2 \underbrace{b_0^\top A_0 x}_{\text{linear}} + \underbrace{b_0^\top b_0}_{\text{constant}}$$

- What is the optimization of constrained linear regression?

Quick Summary

Linear programming (linear objective & constraints):

- **Linear** programming
- **Integer** programming (integer or binary variables)
- **Mixed-integer** programming (mix of continuous & integer variables)

Nonlinear programming (nonlinear objective & constraints):

- A special case: Quadratic programming
- How to reformulate ℓ_2 -norm objective as quadratic programming?